www.csvtuonline.com

Printed Pages – 4

Roll No. : .....

337652(37)

B. E. (Sixth Semester) Examination, Nov.-Dec., 2015

(New Scheme)

(Mechanical Engg. Branch)

ENERGY SYSTEMS

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt all the questions. In each question Part (a) is compulsory and attempt any two from (b), (c) and (d).

Unit-I

1. (a) Explain the basic principle of jet propulsion. 2

(b) Briefly explain ramjet engine with suitable schematic and pressure velocity diagram. 7

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(c) A turbojet engines takes 50 kg/sec of air and propels an aircraft uniform flight speed of 880 km/hr. isentropic enthalpy change for nozzle is 188·28 kJ/kg and its velocity coefficient is 0·96. The fuel/air ratio is 1·2% combusion efficiency 95%, LCV of fuel is $4{\cdot}393 \times 10^4$ kJ/kg.

Find out :

(i) Fuel flow in kg/hr and specific fuel consumption.

(ii) Thermal efficiency of the engine.

(iii) Thrust power.

(iv) Propulsive efficiency

(v) Overall efficiency 7

(d) Draw the sketch and explain, a gas turbine power plant for air craft propulsion. 7

Unit-II

2. (a) Why a rocket is called non-air breathing engine? 2

(b) Draw the schematic of solid propellant rocket and explain its working with its advantages and disadvantages. 7

www.csvtuonline.com

www.csvtuonline.com

(c) Describe gas pressure feed system for liquid propellant rocket engines and mention advantages and dis-advantags rocket engine. 7

(d) A missle has a maximum flight speed to jet speed ratio of 0·2105 and specific impulse equal to 203·88 seconds. Determine for a burn out time of 8 seconds. 7

(i) Effective jet velocity.

(ii) Mass ratio and propellant mass fraction.

(iii) Maximum flight speed.

(iv) Attitude gain during powered and coasting flights. 7

Unit-III

3. (a) What is Solar Energy? Give at least two application of solar energy. 2

(b) Describe in detail a liquid flat-plat collector. How will you increase its efficiency? 7

(c) Draw the schematic of solar absorption refrigeration system and write briefly about it. 7

(d) Calculate the angle made by the beam radiation with the normal to a flat plate collector, pointing due south located in New Delhi (28° 38′N 17′ E) at 9 : 00 hour,

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

solar time on December 1. The collector is titled at angle of 36° with the horizontal.

Unit-IV

4. (a) Explain the term "Photosynthesis".

(b) What is meant by anaerobic digestion?

(c) Prove that the maximum power which can be extracted from wind is 59·3%.

(d) Explain the working of horizontal axis Two-Blade wind mill with suitable diagrams.

Unit-V

5. (a) Explain the fuel cell with classification. 2

(b) Briefly explain various methods of utilization of tidal energy. 7

(c) Describe the closed cycle MHD system with a help of neat sketch. 7

(d) Explain the working of vapour dominated geothermal power plant with help of suitable diagram. 7

www.csvtuonline.com

www.csvtuonline.com